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Flexible and robust piezoelectric polymer nanocomposites based energy harvesters

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Key words: Nanocomposite, Energy harvester, Piezoelectricity, P(VDF-TrFE), Flexible.

Abstract:

Environment friendly, flexible and robust sensors have attracted considerable research attention due to their potential for a wide range of devices in energy generation and harvesting, sensing and biomedical applications. In this manuscript, we demonstrate a lead-free, solution processed flexible piezoelectric energy generator based on a nanocomposite film, consisting of MgO nanoparticles of sizes around <50 nm, embedded in P(VDF-TrFE) matrix. Piezoelectric, ferroelectric and leakage current measurements made on samples with various concentrations of MgO nanoparticles revealed a dramatic improvement in these characteristics at 2 wt% MgO with nearly 50% increase in the piezoelectric response as compared to pure P(VDF-TrFE), attributed to the preferred conformation of PVDF-TrFE chain, improved crystallinity of the P(VDF-TrFE) matrix and uniform distribution of nanoparticles. As assessment of the interactions between –OH groups attached to MgO surface and P(VDF-TrFE), carried out using Fourier-transform infrared spectroscopy (FTIR), suggested weak van der Waals forces between –OH groups and P(VDF-TrFE) being responsible for the observed improvement. This flexible nanocomposite device exhibits superior energy harvesting performance with over two-times improvement in the voltage output (2 V) compared to device using P(VDF-TrFE) films alone. Along with superior electrical properties, nanocomposites also exhibit excellent endurance against electrical as well as mechanical fatigue, with piezoelectric coefficient remaining unchanged even after 10,000 bending cycles, supporting their suitability in flexible energy harvesting applications.

Introduction:

Over the past decade, ferroelectric poly(vinylidene difluoride) [P(VDF)] and its copolymer with trifluoroethylene i.e. P(VDF-TrFE) have been of substantial interest in sensors¹, actuators² and memories³. The material has also gained significant attention in the field of flexible electronic devices owing to its low-cost, low-temperature processing and mechanical flexibility. PVDF occurs in several crystalline forms such as α , β , γ and δ ⁴. Among these, α -phase is a thermodynamically favorable phase but is non-polar and for ferroelectricity to prevail, one needs to convert the α -phase into polar β -form⁵. High degree of crystallinity is required to get polar β -phase that is responsible for better ferro- and piezoelectric response. Previous studies have suggested that various processing conditions such as mechanical stretching, poling or annealing are required to improve the crystallinity of the material⁶. In addition, the formation of polar phase of PVDF can be enhanced by addition of nanosized insulating fillers of ZnO⁷, BaTiO₃⁸, and Al₂O₃⁹ as well as addition of conductive materials such as graphene¹⁰ and silver nanoparticles¹¹. Fillers such as BaTiO₃ possess high dielectric constant and low breakdown strength¹². Large differences between the dielectric constant values of ceramic fillers and the polymer matrix were shown to result in highly inhomogeneous and localized electric fields in the composites which further reduced the breakdown strength of the composites¹³. On the other hand, use of conductive fillers such as silver in large concentrations increases the leakage current by creating leakage paths for charge conduction¹¹. For example, silver based nanocomposites show improved piezoelectricity but at the expense of other electrical characteristics that are desirable for a high-performance device¹¹.

For applications of P(VDF-TrFE) relating to energy harvesting or generation, not only one needs to enhance its piezoelectric characteristics as the generated power is proportional to the piezoelectric coefficient¹⁴, reduced electrical leakage is also important. For P(VDF-TrFE), literature reports piezoelectric coefficient values of the order of 20-40 pC/N^{15,16} that are significantly lower than those of inorganic materials such as Pb(Zr,Ti)O₃ (PZT) which show d_{33} values as high as *ca.* 400 pC/N¹⁷. Hence, even though organic P(VDF-TrFE) are relatively environmentally benign, their lower d_{33} necessitates use of PZT and similar compounds despite disadvantages such as toxicity as well as higher processing temperatures. Therefore, if ferroelectric and piezoelectric properties of P(VDF-TrFE) can further be improved along with modified and easier processing, it could pave the way for piezoelectric P(VDF-TrFE) based flexible energy generation or energy harvesting devices. In the quest for a high performance nanocomposite, we chose to use MgO nanoparticles as fillers in P(VDF-TrFE) because its

dielectric constant is ~ 10 which is very similar to P(VDF-TrFE)¹⁸ and hence, incorporation of MgO nanoparticles is unlikely to result in any inhomogeneity in the electric field distribution in the composite and hence it could be appropriate as a filler material. Moreover, beneficial role of MgO in increasing the breakdown strength¹⁸⁻²⁰ and imparting low leakage current²¹ is already reported. Hence, in addition to improving its ferroelectric and piezoelectric properties, it is expected that MgO fillers will also improve the long-lasting issue of large leakage current of P(VDF-TrFE).

In this manuscript, we report the fabrication of light weight and flexible MgO/P(VDF-TrFE) nanocomposites using a simple solution casting method. Our results show considerable improvement in the ferroelectric and piezoelectric characteristics along with electrical properties such as leakage current and electrical fatigue in MgO doped nanocomposites. Nearly 50% improvement in the piezoelectric response of nanocomposite is observed in comparison to pure P(VDF-TrFE). It is noteworthy that in our work, no surface modification of MgO nano fillers has been done, yet very small quantity of MgO is capable of substantial improvement. Finally, the work shows the utility of MgO-P(VDF-TrFE) nanocomposites for piezoelectric energy generation with excellent resistance to mechanical fatigue.

Experimental details

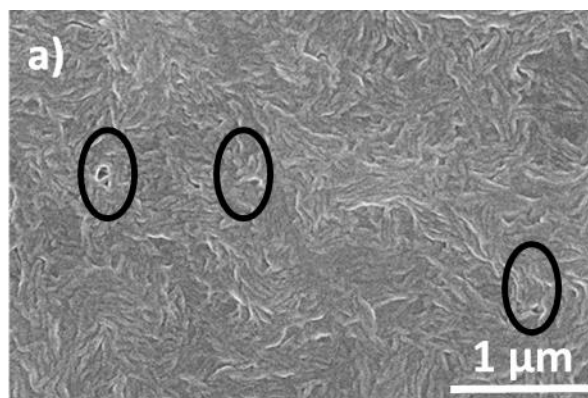
To prepare MgO/P(VDF-TrFE) solution, MgO nanoparticles (sigma Aldrich, particle size < 50 nm) were weighed corresponding to 2, 4, 6 and 8 wt% and were pre-dispersed in 3 ml of methyl ethyl ketone (MEK) by ultrasonication for 30 minutes followed by addition of P(VDF-TrFE) (70:30 mol%, Piezotech, France) to this solution. Each solution was kept on a hot plate for 6 hours with a mechanical mixing at 60°C. The well-mixed solutions were casted onto a glass plate followed by slow evaporation of the solvent by curing the samples at 70°C for 30 min on a hot plate. After drying, the nanocomposite film was cured at 140°C for 1 h followed by ice water quenching⁵ and then peeled off from the glass substrate to achieve a flat and uniform film of MgO/PVDF-TrFE composite (thickness 20 ± 2 μm measured by Mitutoyo 293-831-30 digital micrometer, also verified by cross section SEM analysis, as shown in the supplementary information, section S1, Fig. S1). Free standing films were cut into sizes of 1×1 cm^2 and to perform electrical measurements, top and bottom electrodes of size 5×5 mm^2 were formed by applying silver paste. Ferroelectric and electric fatigue measurements were performed using Radiant Precision Premier II ferroelectric tester. Nova NanoSEM 450 was used for microstructural

characterization. Phase analysis was conducted on a Panalytical MRD Pro X-ray diffractometer whilst Piezoelectric measurements on free standing films were conducted using an AixACCT measurement system. To record the piezoelectric response, Keithley 6517A electrometer (input impedance $200\text{T}\Omega$) was used. Fourier transform infrared spectroscopy (FTIR) measurements were performed using an Agilent technologies Cary 600 series, FTIR spectrometer. Morphology of the films was measured using Asylum Research MFP-3D AFM, as shown in the supplementary information.

Results and Discussion

(a) Structure and Morphology Analysis

For organic nanocomposites, uniform dispersion of inorganic nanoparticles in the organic matrix is highly desirable to have a significant effect on the properties of composite materials. This was assessed by microstructural examination of P(VDF-TrFE) films, embedded with MgO nanoparticles, with 0 to 8 wt% and the corresponding images are shown in Fig. 1. The micrographs depict a dense needle-shaped flaky structure of the surface morphology of pure P(VDF-TrFE) films, as also reported previously⁵. In the MgO/P(VDF-TrFE) nanocomposites, MgO disperses homogeneously in the polymer matrix except at 8wt% concentration where certain level of agglomeration of MgO particles is observed (also verified by the AFM images provided in the supplementary information section S2.1, Fig. S2). SEM images confirm that as compared to the pure P(VDF-TrFE) films, MgO/P(VDF-TrFE) nanocomposites possess hardly any voids. This indicates that the interaction of MgO with P(VDF-TrFE) polymer chains improves surface morphology as well as microstructural integrity.



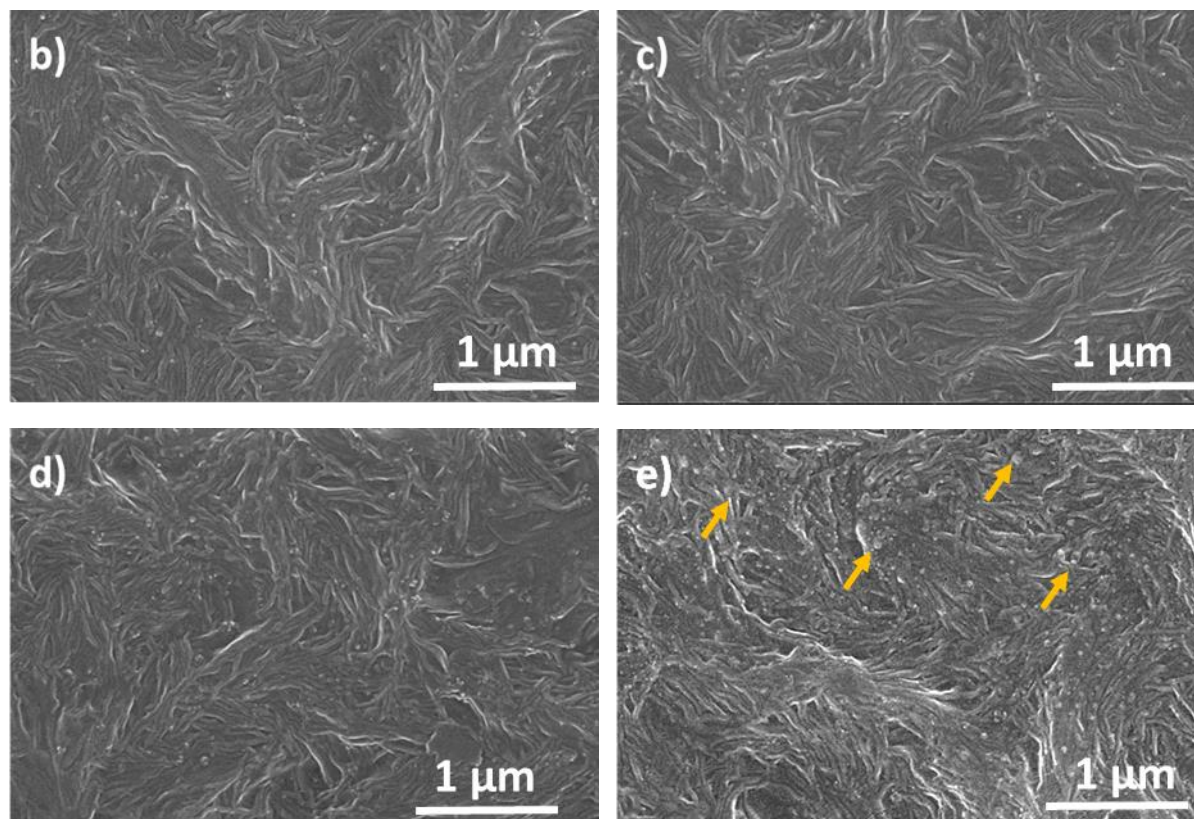


Figure 1: SEM images of (a) 0wt%, (b) 2wt%, (c) 4wt%, (d) 6wt% and (e) 8wt% MgO in P(VDF-TrFE). Arrows are guide to eye to show the agglomeration of MgO at higher MgO loading.

Next, we conducted FTIR on these nanocomposite samples to analyze the phases present in the films. Fig. 2 shows the FTIR spectra of PVDF-TrFE/MgO composite films with various wt% of MgO. In the spectrum of MgO nanoparticles (Fig. 2(a)), broader absorption peaks were detected at 1440 and 3360 cm^{-1} that signify the presence of MgO and $-\text{OH}$ group respectively²². On the other hand, the fingerprints of polar β -phase are evident by the presence of well-defined peaks at 885 cm^{-1} and 1288 cm^{-1} in pure P(VDF-TrFE) film FTIR spectra (Fig. 2(b))²³. In the composite samples, the band at 1173 cm^{-1} is attributed to $-\text{CF}_2$ stretching mode of P(VDF-TrFE) and exhibits a noticeable shift in position from 1173 cm^{-1} to 1169 cm^{-1} when loaded with MgO (Fig. 2 (c)). This is due to MgO/ CF_2 interactions which lead to a reduction in the effective mass of CF_2 dipoles and hence a shift towards lower wavenumber side is observed signifying a reduction in the vibrational frequency of the CF_2 band. Moreover, additional peaks at 2975 and 3010 cm^{-1} (Fig. 2 (d)) evolve, particularly in 2% and 4% MgO samples. Although exact reason is not clear, these could be due to the presence of CH/CH_2 bond²⁴.

Hence, change in the peak positions and presence of new absorption peaks suggest towards a chemical interaction between P(VDF-TrFE) and MgO.

Further, to confirm whether MgO loading has any effect on crystallinity of the composite, we investigated the X-Ray diffraction (XRD) spectra of the samples. The XRD patterns, as shown in Fig. 3 (a), illustrate that the addition of MgO nanoparticles has a strong effect on the orientation of chains in P(VDF-TrFE) as shown by the changes in the intensity of the peak around $2\theta \sim 20^\circ$, attributed to (110) and (200) planes²⁵. Incorporation up to 2 wt% MgO enhances the crystallinity which gradually decreases with further addition of MgO nanoparticles, as shown in the inset of Fig. 3(a). The percentage of crystallinity (χ_c) of the nanocomposites was calculated using the equation:

$$\chi_c(\%) = \frac{\sum A_c}{\sum A_c + \sum A_{am}} \times 100$$

Where $\sum A_c$ is total area of crystalline zone and $\sum A_{am}$ is total area of amorphous zone. The peak at *ca.* 44° in the XRD spectra as shown in Fig.3 (b) is attributed to MgO.

The possible reason behind initially improved crystallinity of nanocomposites is due to formation of hydrogen bond between the “inherently present -OH group on the MgO surface”²⁶⁻²⁸ and F atom of P(VDF-TrFE) [29]. This is also evident from the FTIR spectra where the peak corresponds to CF₂ stretching shifted from 1173 cm^{-1} to 1169 cm^{-1} . This interaction between MgO nanoparticles and the P(VDF-TrFE) chains increase the crystallinity of the films. However, as the MgO content is increased, the mobility of the chains will get significantly restricted because of hydrogen bond formation rendering more stiffness in the chains. Hence, as a result, at higher concentrations, MgO fillers act as an obstacle that prevents the organization of polymer chains during annealing which, in turn, results in lowering of crystallinity. To get more insight into it, we also fabricated free standing PVDF films (less polar than P(VDF-TrFE)) and 2wt% MgO/PVDF films. Fig. 3 (c) depicts the comparison of XRD of both the films. The intensity of (020) peak, attributed to α -phase, reduces with 2 wt% loading of MgO. This corroborates that MgO enhances the crystallinity of the films and stabilizes the β -phase of PVDF. Fig. 4 schematically shows the interaction of -OH group present on the surface of MgO nanoparticles with F atoms of P(VDF-TrFE) chains. The interaction between these two results in the formation of hydrogen bond which is perhaps the reason behind enhanced crystallinity. Similar behavior is previously reported with surface hydroxylated BaTiO₃ nanoparticles^{24,29} in which hydroxyl groups attached to the BaTiO₃ increased the charge accumulation on its surfaces leading to increased polarization.

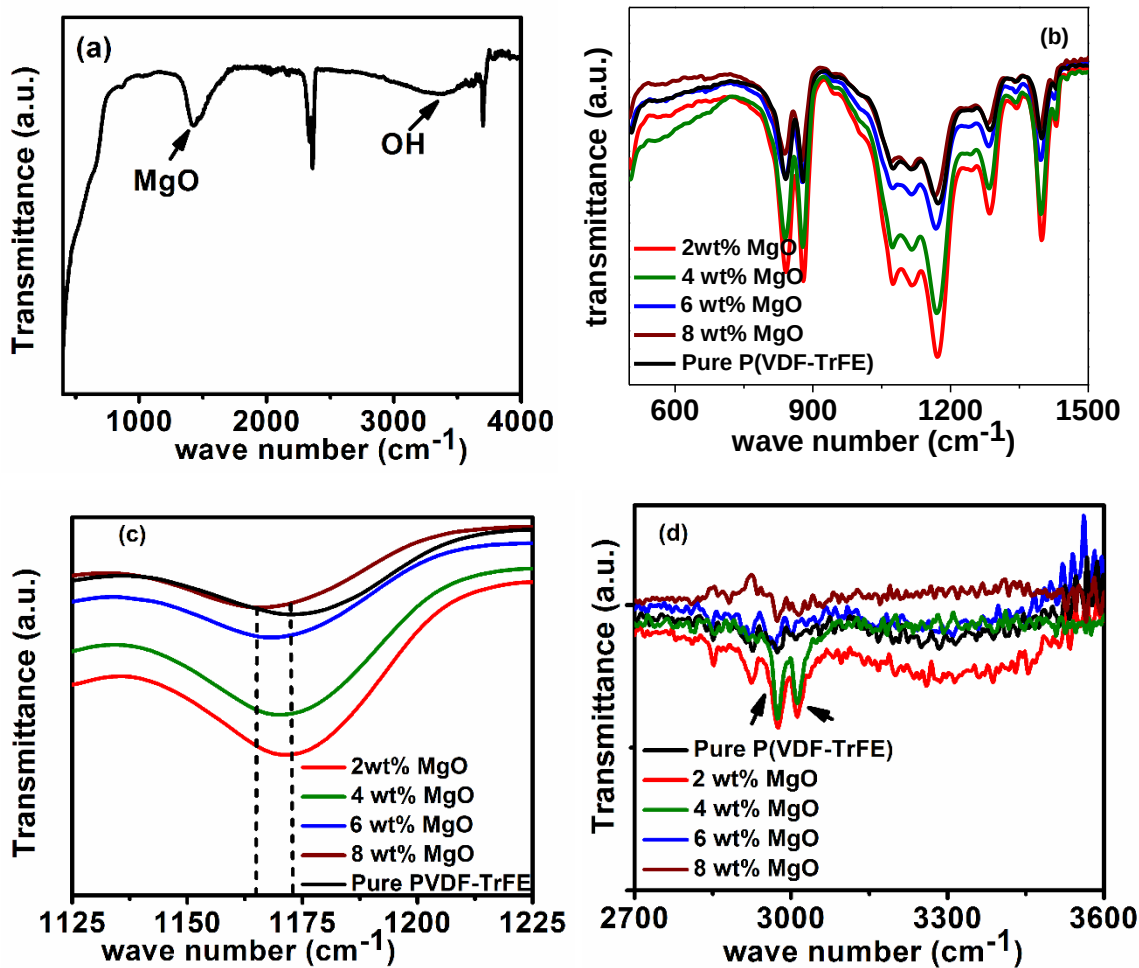
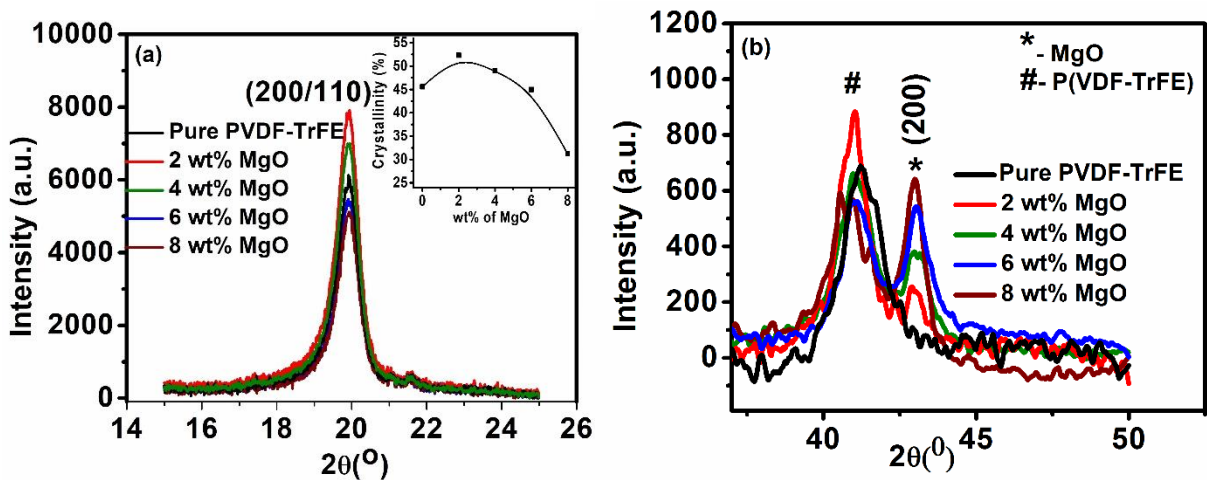


Figure 2: FTIR spectra of (a) pure MgO, (b) (c) and (d) show FTIR spectra of MgO/P(VDF-TrFE) nanocomposite films with different MgO loading levels in different wave number ranges (b) 500-1500 cm^{-1} , (c) 1125-1225 cm^{-1} and (d) 2700-3600 cm^{-1} .



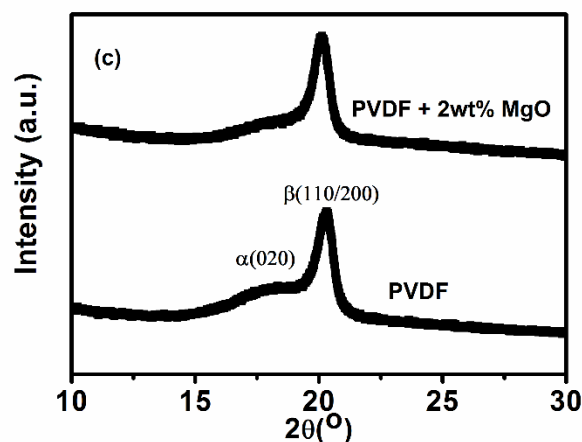


Figure. 3: (a) XRD spectra ($2\theta = 15^\circ$ to 25°) to confirm the β phase of P(VDF-TrFE) at $2\theta = 19.7^\circ$, (b) XRD spectra in the range $2\theta = 35^\circ$ to 50° showing MgO (200) peak and (c) comparative XRD spectra of PVDF films with 2 wt% MgO and without MgO. Inset of (a) shows variation of %crystallinity of the P(VDF-TrFE) films with various MgO loading levels.

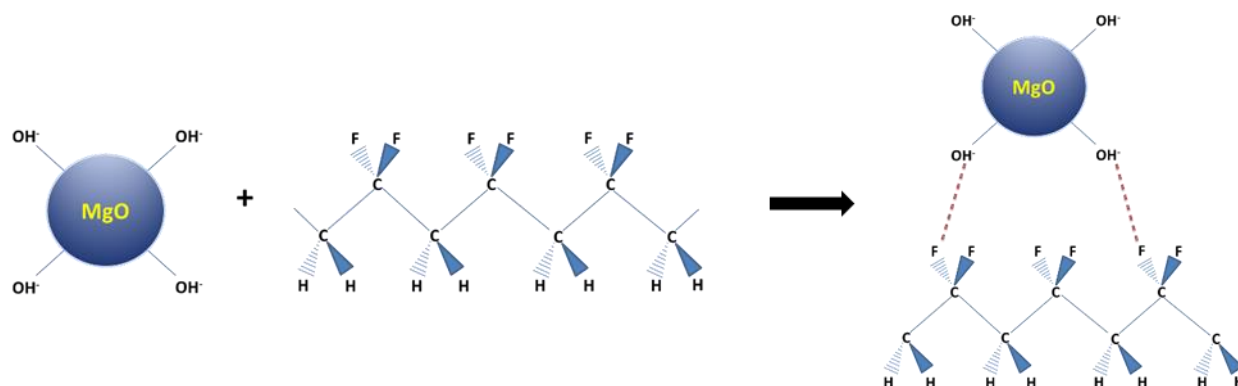


Figure: 4: Schematic diagram showing formation of hydrogen bond in MgO/P(VDF-TrFE) nanocomposites

(b) Ferroelectric and Piezoelectric Properties

Next, we analyzed the effect of MgO loading on the electrical properties of P(VDF-TrFE) thin films loaded with MgO nanoparticles. Polarization versus electric field curves of P(VDF-TrFE) films with various MgO loading levels are shown in Fig. 5(a-b). We see that both saturation and remanent polarization values first increase at MgO level of 2 wt% and then start decreasing upon further increasing MgO content. In contrast, reverse trend is observed in the coercive field values of the capacitors. The reason behind reduction in the coercive field may be related to the affinity between P(VDF-TrFE) and MgO. Poor affinity between hydrophobic P(VDF-TrFE) and hydrophilic MgO could reduce the glass transition temperature and improve the chain mobility³⁰. Increased chain mobility may trigger fast polarization switching by reducing the energy barrier needed for the switching and hence low coercive field was observed. However, as the amount of MgO loading increases, nanoparticles may

act as an obstacle for the chain movement especially with 8wt% MgO loading. The agglomeration observed with 8wt% MgO loading acts a defect in the polymer matrix and hence these devices end up with substantially lower polarization as well as higher coercive field. The observed trend in ferroelectric polarization is consistent with the observations of XRD and FTIR data owing to the higher crystallinity of nanocomposites of 2wt% MgO. We also examined the nanocomposites with 1wt% and 3wt% MgO to ascertain the exact amount of the filler needed for best device performance (results not shown here). It is observed that composites with 3wt% MgO yield similar polarization values as with 2wt% MgO while 1wt% MgO devices has less polarization values than 2 wt% MgO loading. Hence, 2wt% MgO is the ascertained as optimal level of loading for the highest ferroelectric response of nanocomposites films.

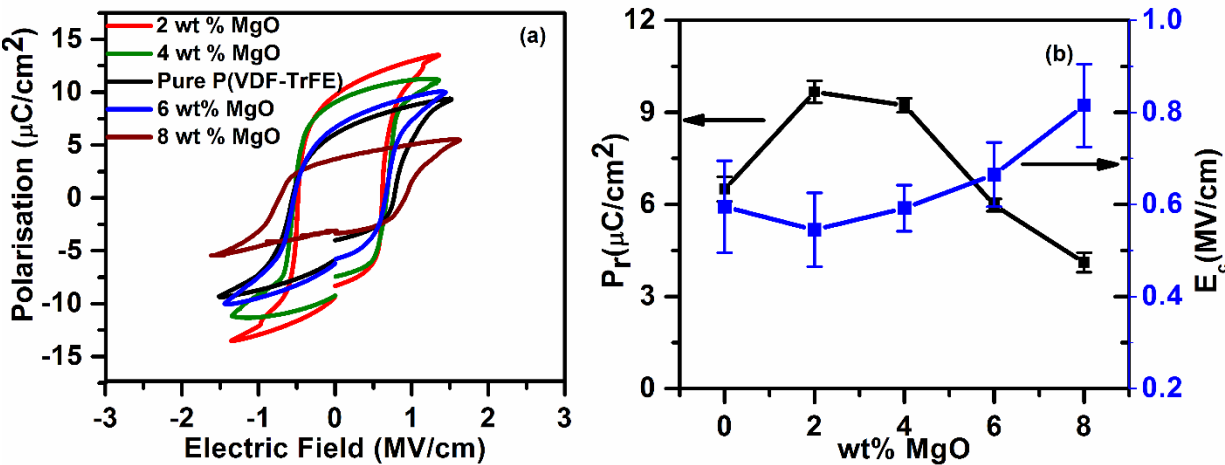


Figure.5: (a) Polarization-Electric field hysteresis curves and (b) P_r and E_c values of MgO/P(VDF-TrFE) nanocomposites with different MgO loading.

P(VDF-TrFE) based capacitors, despite having good ferroelectric properties, exhibit higher electrical leakage (acceptable values are in the range of 10^{-6} A/cm²) and low electrical fatigue endurance, preventing them to fulfil their application potential. Since, MgO is known to have very low electrical conductivity, addition of MgO filler to P(VDF-TrFE) should improve its electrical resistance. Fig. 6(a) illustrates the comparison of leakage current values of ferroelectric polymer with and without MgO nanoparticles. The leakage current density of pure P(VDF-TrFE) based devices are approximately 10^{-5} A/cm² at electric field of 1 MV/cm, comparable to the reported values³¹. In contrast, the leakage currents of the composites with 2 wt% of MgO/P(VDF-TrFE) composites are lowered by an order of magnitude, suggesting beneficial role of MgO in enhancing the electrical insulation of the composite film. Furthermore, MgO addition also provides great endurance against electrical fatigue as depicted in

Fig. 6 (b) which shows only 10% degradation in the polarization after 10^6 bipolar switching cycles while pure P(VDF-TrFE) based devices show 50% decrease in polarization in the same period. This remarkably improved fatigue response is attributed to interactions between MgO nanoparticles and P(VDF-TrFE) which results in improved chain stability and enhanced crystallinity of P(VDF-TrFE), reduced gauche effects (e.g. chain kinks and entanglement), leading to suppression of defluorination which is a major cause behind electrical fatigue²³.

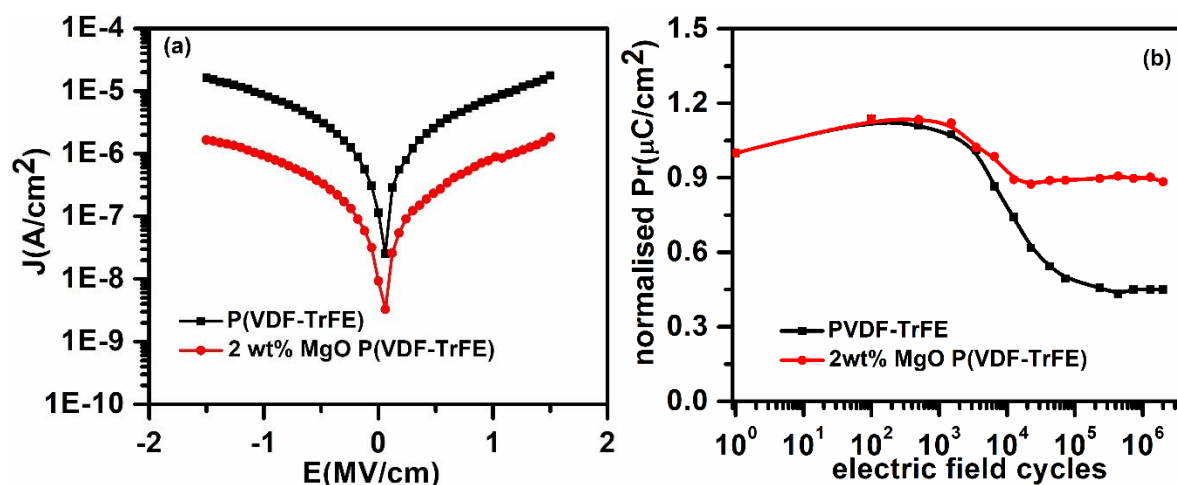


Figure 6: (a) Leakage characteristics and (b) electric fatigue endurance of pure P(VDF-TrFE) films and those with 2wt% MgO loading.

We then evaluated the piezoelectric characteristics which are also strongly dependent on the crystallinity of the polymer and polar phase development in the nanocomposite. Fig. 7 shows that the piezoelectric coefficient (d_{33}) values depict a similar trend as of remanent polarization with highest value (~ 65 pm/V) obtained for 2 wt% MgO/P(VDF-TrFE) nanocomposite film. In contrast, devices with pure P(VDF-TrFE) exhibit d_{33} value of ~ -40 pm/V, comparable to the previously reported values³ showing that addition of 2wt% MgO nanoparticles in P(VDF-TrFE) films leads to almost 45% improvement in piezoelectric coefficient, confirming the efficacy of MgO addition in P(VDF-TrFE) matrix to achieve superior ferroelectric and piezoelectric response. The piezoelectric coefficient values obtained in this work are superior than the previously reported values of P(VDF-TrFE) loaded with other nanoparticles such as Al₂O₃⁹, PZT³², 65PMN-35PT³³, ZnO³⁴ and Ag¹¹.

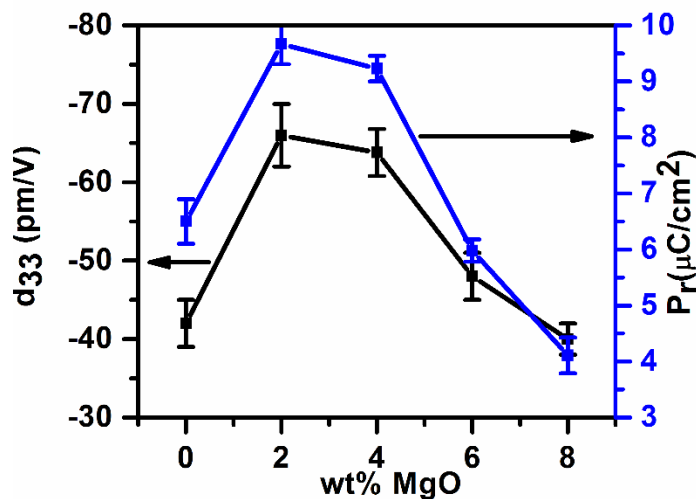


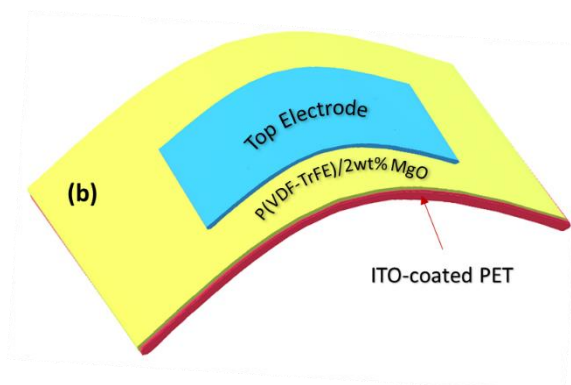
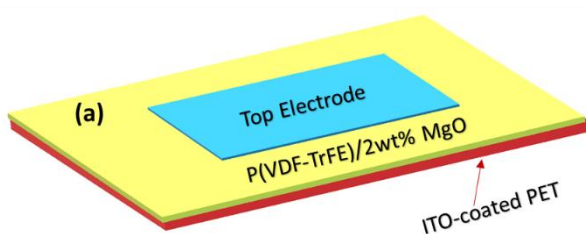
Figure 7: Variation of piezoelectric coefficient of MgO/P(VDF-TrFE) nanocomposite films with different MgO loading levels.

(c) Piezoelectric Energy Harvesting on Flexible Substrates

Fig. 8 (a-b) show the images of the flexible and bendable P(VDF-TrFE) based power generator which was fabricated by casting the composite films on ITO coated PET substrates (substrate thickness: 125 μm). ITO films on PET substrates acted as the bottom electrode over which 2wt% MgO/P(VDF-TrFE) of thickness 20 μm was deposited. For the top electrode, silver paste was applied on the area of 7x7 mm^2 . Thus, MIM capacitor of area 49 mm^2 and thickness 20 μm was achieved. Measurements of piezoelectric response were repeated over 10 devices (details are provided in supplementary information S2.3). The device can be considered as a beam within the framework of continuous medium elasticity theory for 'classical' bending geometry³⁵. In this configuration, strain experienced by the lower (inner) surface of the PET substrate is compressive whilst the upper (external) surface experiences tensile stress. A neutral plane, where the strain is zero, lies in the middle and is likely to lie on the substrate itself given that film is substantially thinner (20 μm) than the substrate (125 μm). Therefore, the MgO/P(VDF-TrFE) nanocomposite films, deposited on top of the upper(external) surface of the substrate is likely to experience a tensile stress. We employed a cyclic strain to this flexible device using a custom-built setup as demonstrated in our previous work²³. Devices were bent and flattened at a bending radius of 6 mm while the bending frequency was maintained at 1 Hz and the output voltage was measured. As results in Fig. 8 (c-d) show, the 2 wt% MgO/P(VDF-TrFE) nanocomposite devices demonstrate superior piezoelectric properties than pure P(VDF-TrFE), attributed to higher amount of polar β -phase in nanocomposite. The nanocomposite devices show an output voltage of 2V which is more than double

of that exhibited by pure P(VDF-TrFE) devices (0.9 V). Applying strain to the MgO/PVDF-TrFE composite films thus produces deformation in its crystal structure which induce charges causing a piezopotential across it. To compensate this potential, external charges will flow to the capacitor resulting in an output signal. When pressure is removed, piezoelectric potential between the electrodes also diminishes. Free charges flow back owing to an electric pulse in the opposite direction. Thus, the positive and the negative sign of the voltage amplitude signifies imparting (bending) and releasing (flattening) of the pressure, respectively, as depicted by Fig. 8 (c-d).

Most common forms of mechanical energy around us which otherwise go to waste can be derived from human motion-based activity. This kind of harvesting of mechanical energy into electrical energy is a useful facet of piezoelectric materials. In the context of ferroelectric polymers such as P(VDF-TrFE), it can be quite useful for various flexible and wearable devices. For a practical demonstration of such touch based devices, we applied pressure on 2 wt% MgO/P(VDF-TrFE) nanocomposite film-based power generator by tapping with a finger. The output voltage generated by tapping is shown in Fig. 9 (a) which generates a voltage of *ca.* 0.4 V upon repeated touching with remarkable consistency. This illustrates that these MgO/P(VDF-TrFE) nanocomposite based films can be used in wearable devices such as touch sensors and other self-wearable devices based on harvesting of mechanical energy.



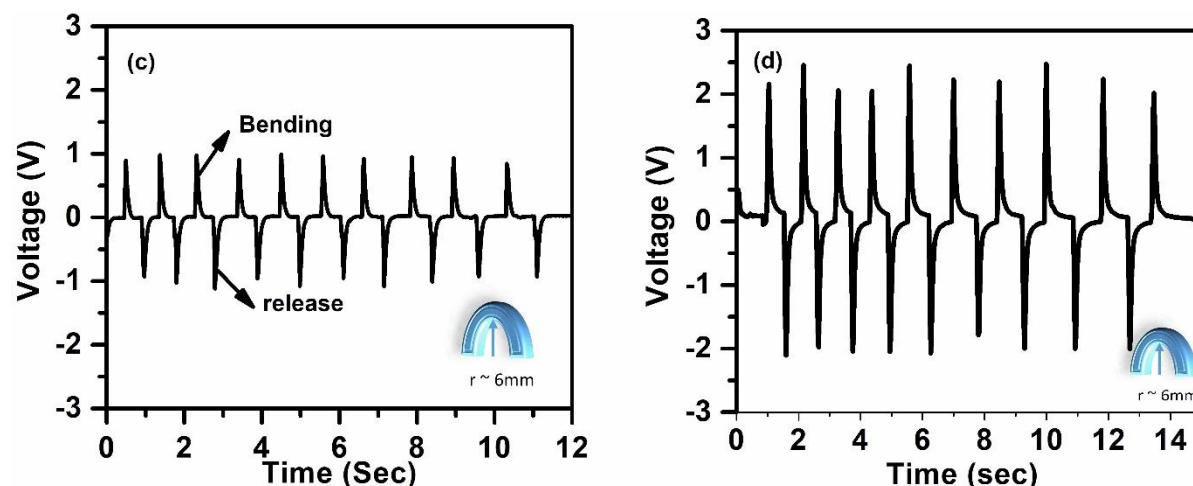


Figure 8: Image of 2 wt% MgO/ P(VDF-TrFE) in (a) release and (b) bend mode, demonstrating the device flexibility. Plots of open-circuit voltage generated from (c) pure P(VDF-TrFE) and (d) 2 wt% MgO/ P(VDF-TrFE) power generators under the cyclic bending and releasing motions with 6 mm bending radius and at approximately 1 Hz frequency.

For long term operation of these devices, one requires the devices not only to be flexible but also robust at the same time. Hence, to demonstrate the robustness of the composite films in practical applications where a device experiences multiple strain cycles before it is discarded, we applied multiple strain cycles to the flexible power generators. Bending test was performed by placing the composite films between two rods in which the bottom rod is fixed while top rod is moved in a fashion as shown in Ref. 23. A fixed number of bending cycles were applied on the devices by rotating the top rod at a fixed speed. The bending speed of the set-up can be changed through the motor while bending radius is set by positioning the lower rod. We carried out the tests of mechanical strain cycling at bending speeds of 20, 50 and 100 cycles per minute with bending radius of 6mm. Fig. 9 (b) shows the effect of multiple bending cycles on the d_{33} values and the results suggests that after 10,000 bending cycles, the d_{33} values remain unaffected irrespective of the bending speed, confirming the robustness of the composite devices and that these can be used in practical applications without any mechanical degradation.

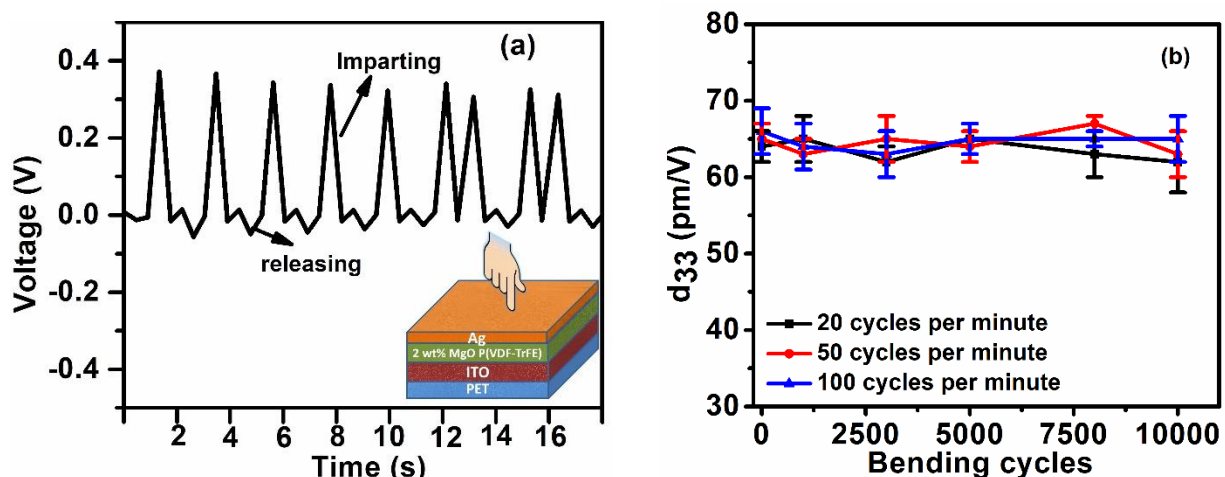


Figure 9: (a) Voltage generation by finger tapping of the 2 wt% MgO/P(VDF-TrFE) nanocomposite device and (b) variation of piezoelectric coefficient as a function of bending cycles at various strain cycling rates.

Conclusions

In this work, we fabricated MgO/P(VDF-TrFE) nanocomposite films by solution casting method which were successfully demonstrated as piezoelectric energy generating devices with excellent mechanical robustness. Devices with 2 wt% MgO in P(VDF-TrFE) showed peak ferroelectric and piezoelectric properties with almost 45 % improvement in polarization and piezoelectric response. This improvement emanates without sacrificing any other electrical properties. In addition, nanocomposite also exhibit an order of magnitude decrease in the electrical leakage current and remarkable improvement in electrical fatigue endurance. The improvement in the electrical properties of nanocomposite films is attributed to improved crystallinity due to chemical interactions between MgO nanoparticles and P(VDF-TrFE). Further testing of devices as piezoelectric energy generator on flexible substrates showed an output voltage of 2 V in nanocomposite devices which was more than twice to that of pure P(VDF-TrFE) films. The devices also exhibited excellent response to the touch showing its potential as a touch sensor. It also exhibited excellent mechanical robustness when they were subjected to mechanical strain cycling for up to 10,000 cycles. This work clearly shows the potential of MgO/P(VDF-TrFE) nanocomposites as excellent ferroelectric and piezoelectric materials which could be useful for energy harvesting devices as well as touch sensors.

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References

1. Khan, S.; Dang, W.; Lorenzelli, L.; Dahiya, R. Flexible pressure sensors based on screen-printed P(VDF-TrFE) and P(VDF-TrFE)/MWCNTs. *IEEE Trans. Semicond. Manuf.* **2015**, *28*, 486–493.
2. Xu, T. B.; Cheng, Z. Y.; Zhang, Q. M. High-performance micromachined unimorph actuators based on electrostrictive poly(vinylidene fluoride-trifluoroethylene) copolymer. *Appl. Phys. Lett.* **2002**, *80*, 1082–1084.
3. Singh, D.; Deepak; Garg, A. An efficient route to fabricate fatigue-free P (VDF-TrFE) capacitors with enhanced piezoelectric and ferroelectric properties and excellent thermal stability for sensing and memory applications. *Phys. Chem. Chem. Phys.* **2017**, *19*, 7743–7750.
4. Jurczuk, K.; Galeski, A.; Mackey, M.; Hiltner, A.; Baer, E. Orientation of PVDF α and γ crystals in nanolayered films. *Colloid. Polym. Sci.* **2015**, *293*, 1289–1297.
5. Singh, D.; Garg, A.; Deepak. Cooling rate controlled microstructure evolution and reduced coercivity in P (VDF-TrFE) devices for memory applications. *Org. Electron.* **2014**, *15*, 82–90.
6. Ling, Q.-D.; Liaw, D.-J.; Zhu, C.; Chan, D. S.-H.; Kang, E.-T.; Neoh, K.-G. Polymer electronic memories: Materials, devices and mechanisms. *Prog. Polym. Sci.* **2008**, *33*, 917–978.
7. Li, Z.; Zhang, X.; Li, G. In situ ZnO nanowire growth to promote the PVDF piezo phase and the ZnO-PVDF hybrid self-rectified nanogenerator as a touch sensor. *Phys. Chem. Chem. Phys.* **2014**, *16*, 5475–5479.

8. Zhou, T.; Zha, J.-W.; Cui, R.-Y.; Fan, B.-H.; Yuan, J.-K.; Dang, Z.-M. Improving dielectric properties of BaTiO₃ / ferroelectric polymer composites by employing surface hydroxylated BaTiO₃ nanoparticles. *ACS Appl. Mater. Interfaces* **2011**, *3*, 2184–2188.
9. Nguyen, V.-S.; Hadji, R.; Vincent, B.; Rouxel, D.; Sarry, F.; and Bauer, F. P (VDF-TrFE)/Al₂O₃ piezoelectric nanocomposite thin films. IEEE International Symposium on the Applications of Ferroelectrics (ISAF) 2010; 2010;pp 1–4.
10. Achaby, M. E.; Arrakhiz, F. Z.; Vaudreuil, S.; Essassi, E. M.; Qaiss, A. Piezoelectric β -polymorph formation and properties enhancement in graphene oxide-PVDF nanocomposite films. *Appl. Surf. Sci.* **2012**, *258*, 7668–7677.
11. Paik, H.; Choi, Y.-Y.; Hong, S.; No, K. Effect of Ag nanoparticle concentration on the electrical and ferroelectric properties of Ag/P(VDF-TrFE) composite films. *Sci. Rep.* **2015**, *5*, 13209.
12. Wang, Z.; Nelson, J. K.; Hillborg, H.; Zhao, S.; Schadler, L. S. Dielectric constant and breakdown strength of polymer composites with high aspect ratio fillers studied by finite element models. *Compos. Sci. Technol.* **2013**, *76*, 29–36.
13. Mazierska, J.; Ledenyov, D.; Jacob, M. V.; Krupka, J. Precise microwave characterization of MgO substrates for HTS circuits with superconducting post dielectric resonator. *Supercond. Sci. Technol.* **2004**, *18*, 18.
14. Song, H.-C.; Kumar, P.; Maurya, D.; Kang, M.-G.; Reynolds, W. T.; Jeong, D.-Y.; Kang, C.-Y.; Priya, S. Ultra-Low Resonant Piezoelectric MEMS Energy Harvester With High Power Density. *J. Microelectromech. Syst.* **2017**, *26*, 1226-1234.
15. Xu, J.; Dapino, M. J.; Gallego-Perez, D.; Hansford, D. Microphone based on polyvinylidene

fluoride (PVDF) micro-pillars and patterned electrodes. *Sens. Actuators, A* **2009**, *153*, 24–32.

16. Oliveira, F.; Leterrier, Y.; M, J.-A.; Sereda, O.; Neels, A.; Dommann, A.; Damjanovic, D. Process influences on the structure, piezoelectric, and gas-barrier properties of PVDF-TrFE copolymer. *J. Polym. Sci., Part B: Polym. Phys.* **2014**, *52*, 496–506.

17. Guo, Q.; Cao, G. Z.; Shen, I. Y. Measurements of piezoelectric coefficient d_{33} of lead zirconate titanate thin films using a mini force hammer. *J. Vib. Acoust.* **2013**, *135*, 011003.

18. Zhou, Y.; He, J.; Hu, J.; Dang, B. Surface-modified MgO nanoparticle enhances the mechanical and direct-current electrical characteristics of polypropylene/polyolefin elastomer nanodielectrics. *J. Appl. Polym. Sci.* **2016**, *133*, 42863.

19. Ju, S.; Zhang, H.; Chen, M.; Zhang, C.; Chen, X.; Zhang, Z. Improved electrical insulating properties of LDPE based nanocomposite: Effect of surface modification of magnesia nanoparticles. *Composites Part A* **2014**, *66*, 183–192.

20. Peng, S.; He, J.; Hu, J.; Huang, X.; Jiang, P. Influence of functionalized MgO nanoparticles on electrical properties of polyethylene nanocomposites. *IEEE Trans. Dielectr. Electr. Insul.* **2015**, *22*, 1512–1519.

21. Lim, M.-H.; Kang, K.; Kim, H.-G.; Kim, I.-D.; Choi, Y.; Tuller, H. L. Low leakage current-stacked Mg O/ $\text{Bi}_{1.5}\text{Zn}_{1.0}\text{Nb}_{1.5}\text{O}_7$ gate insulator—for low voltage ZnO thin film transistors. *Appl. Phys. Lett.* **2006**, *89*, 202908.

22. Li, L.-X.; Xu, D.; Li, X.-Q.; Liu, W.-C.; Jia, Y. Excellent fluoride removal properties of porous hollow MgO microspheres. *New J. Chem.* **2014**, *38*, 5445–5452.

23. Singh, D.; Deepak; Garg, A. The combined effect of mechanical strain and electric field

cycling on the ferroelectric performance of P (VDF-TrFE) thin films on flexible substrates and underlying mechanisms. *Phys. Chem. Chem. Phys.* **2016**, *18*, 29478–29485.

24. Gao, L.; He, J.; Hu, J.; Li, Y. Large enhancement in polarization response and energy storage properties of poly (vinylidene fluoride) by improving the interface effect in nanocomposites. *J. Phys. Chem. C* **2014**, *118*, 831–838.

25. Singh, D.; Deepak; Garg, A. Interface morphology driven control of electrical properties of P (VDF- TrFE) and PMMA blend M-I-M capacitors. *Org. Electron.* **2014**, *15*, 3811–3817.

26. Ciston, J.; Subramanian, A.; Marks, L. D. Water-driven structural evolution of the polar MgO (111) surface: An integrated experimental and theoretical approach. *Phys. Rev. B* **2009**, *79*, 085421.

27. Shido, T.; Asakura, K.; Iwasawa, Y. The hydrogen exchange reaction of surface deuteroxyl groups on MgO with H₂. *J. Chem. Soc. Faraday Trans1.* **1989**, *85*, 441-453

28. Coluccia, S., Marchese, L., Lavagnino, S., & Anpo, M. Hydroxyls on the surface of MgO powders. *Spectrochim. Acta A* **1987**, *43*, 1573-1576.

29. Zhou, T.; Zha, J.-W.; Cui, R.-Y.; Fan, B.-H.; Yuan, J.-K.; Dang, Z.-M. Improving dielectric properties of BaTiO₃/ferroelectric polymer composites by employing surface hydroxylated BaTiO₃ nanoparticles. *ACS Appl. Mater. Interfaces* **2011**, *3*, 2184– 2188.

30. Chu, B. J.; Lin, M. R.; Neese, B.; Zhang, Q. M. Interfaces in Poly(vinylidene fluoride) Terpolymer/ZrO₂ Nanocomposites and their Effect on Dielectric Properties. *J. Appl. Phys.* **2009**, *105*, 014103.

31. Fujisaki, S.; Ishiwara, H.; Fujisaki, Y. Low-voltage operation of ferroelectric poly

(vinylidene fluoride-trifluoroethylene) copolymer capacitors and metal-ferroelectric-insulator-semiconductor diodes. *Appl. Phys. Lett.* **2007**, *90*, 162902.

32. Ng, K. L.; Chan, H. L. W.; Choy, C. L. Piezoelectric and pyroelectric properties of PZT/P (VDF-TrFE) composites with constituent phases poled in parallel or antiparallel directions. *IEEE Trans Ultrason Ferroelectr Freq Control.* **2000**, *47*, 1308-1315.

33. Lam, K. H.; Chan, H. L. W. Piezoelectric and pyroelectric properties of 65PMN- 35PT/P (VDF-TrFE) 0–3 composites. *Compos. Sci. Technol.* **2005**, *65*, 1107-1111.

34. Rouxel, D., Vincent, B., Badie, L., Dos Santos, F. D., Lamouroux, E., & Fort, Y. (2013). Influence of cluster size and surface functionalization of ZnO nanoparticles on the morphology, thermomechanical and piezoelectric properties of P (VDF-TrFE) nanocomposite films. *Appl. Surf. Sci.* **2013**, *279*, 204-211

35. Fukuda, K.; Sekine, T.; Shiwaku, R.; Morimoto, T.; Kumaki, D.; Tokito, S. Free- standing organic transistors and circuits with sub-micron thicknesses. *Sci. Rep.* **2016**, *6*, 27450.

